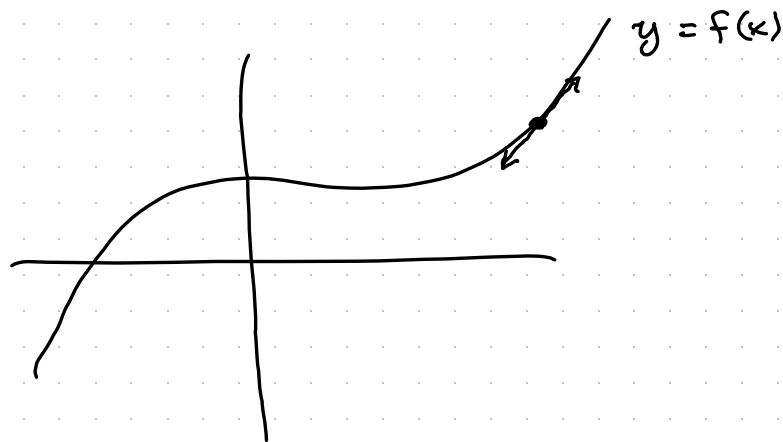
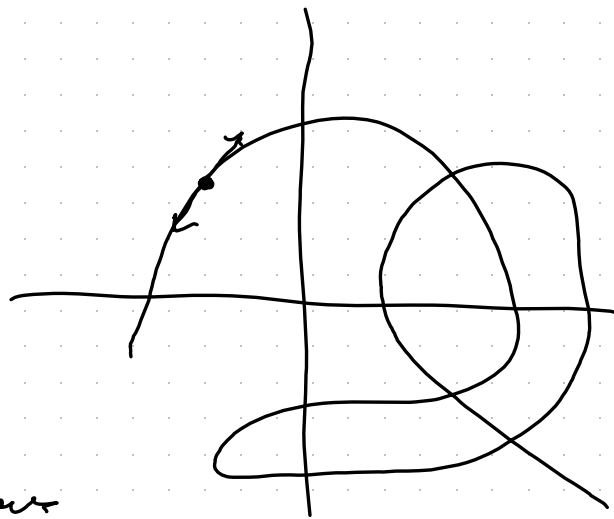


Parametric curves - 7.1



Start thinking of plane curves a little bit more like we do in Calc III, as functions that take 1 input and return 2 outputs.



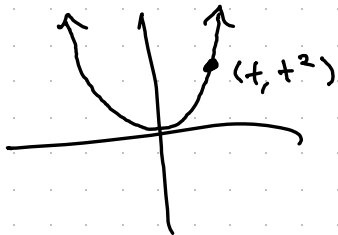
A parametric curve is a function

from the real numbers to the plane.

We think of this real number input often as time and the point output as a position.

We can notate such functions as $(x(t), y(t))$.

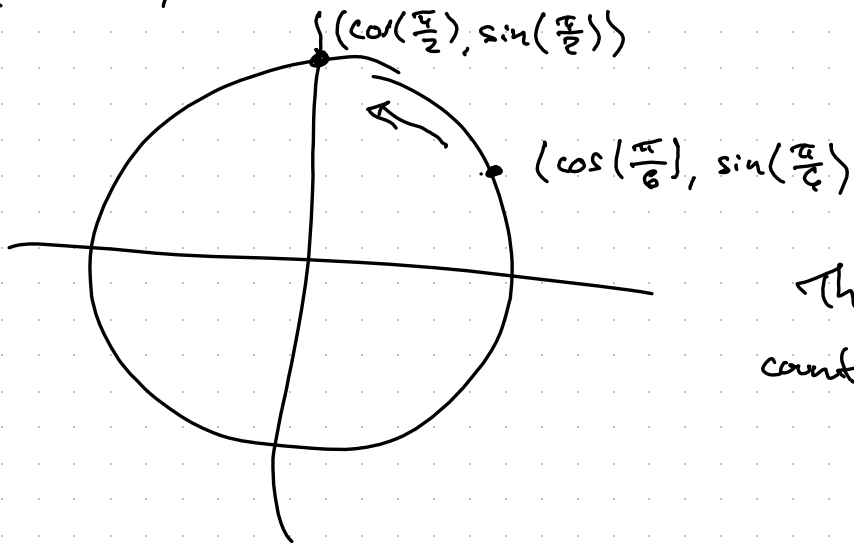
Example: We can parameterize the curve given by $y = x^2$ as (t, t^2) .



(Can always parameterize $y = f(x)$ as $(t, f(t))$ like this.)

Example: We've been describing the unit circle parametrically all along!

If we have a point on the unit circle, we know the coordinates are $(\cos(t), \sin(t))$ for some t .

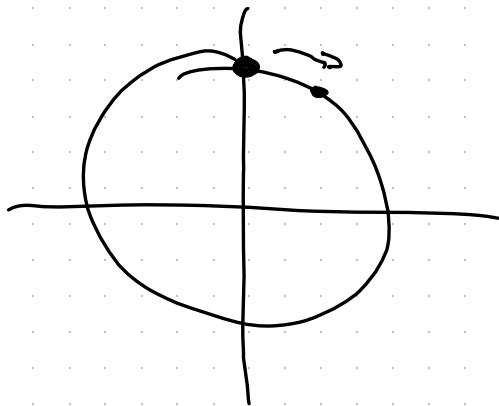


$$x(t) = \cos(t)$$

$$y(t) = \sin(t).$$

This is a
counter-clockwise
parameterization.

Other parameterizations of the circle:



$$(\sin(t), \cos(t))$$

"starts" with $(\sin(0), \cos(0))$
 $= (0, 1)$

Moves clockwise:

$$\left(\sin\left(\frac{\pi}{6}\right), \cos\left(\frac{\pi}{6}\right)\right) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

We could also have obtained a clockwise parameterization by plugging $-t$ for t .

$$\text{That is, } (\cos(-t), \sin(-t)) = (\cos(t), -\sin(t))$$

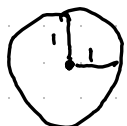
is a clockwise parameterization.

For a circle with radius r , we have $(r\cos(t), r\sin(t))$ as a parameterization.

To center the circle at (a, b) , we can add the functions component-wise:

$$(r\cos(t), r\sin(t)) + (a, b) = (r\cos(t) + a, r\sin(t) + b)$$

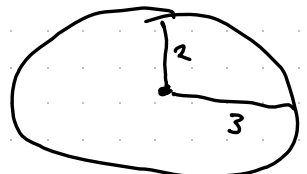
Stretching x and y directions by different amounts, e.g. $(3\cos(t), 2\sin(t))$, gives ellipses.



circle



stretch x



stretch y

Could also center these ellipses at other points.

Example: $(5\sin(t) + 1, 2\cos(t) - 3)$ gives

a clockwise parameterization of an ellipse centered at $(1, -3)$ with major axis 5 and minor axis 2.
(horizontal) (vertical)

Example: If we make the scaling depend on t , we can get something like a spiral:
 $(t\cos(t), t\sin(t))$.